

# JAINDU SAMARANAYAKE

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## EDUCATION

**Bachelor of Science Honors in Electronic and Telecommunication Engineering** March 2024 - Present  
University of Moratuwa, Sri Lanka CGPA: 3.65/4.00

**G.C.E. Advanced Level Examination** 2020 - 2023  
Central College, Kuliyaipitiya Z-Score: 2.1186  
Results: 3 'A' Passes (Combined Mathematics, Physics, Chemistry)

## TECHNICAL SKILLS & INTERESTS

|                                                                 |                                                                                                                                                                           |
|-----------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Hardware &amp; Embedded Engineering Software Programming</b> | ESP32, Raspberry Pi, Arduino Mega, FPGA (DE0-Nano, DE2-115), I2C, UART<br>Altium Designer, SolidWorks, OMNeT++, Cisco Packet Tracer, MATLAB/Simulink<br>C++, Python, Java |
| <b>Domains of Interest</b>                                      | Embedded Systems, Firmware Development, IoT, Network Architecture, ML                                                                                                     |

## HONOURS & AWARDS

- **4th Place** – SPARK Challenge 2026: Recognized for creating sustainable, market-ready IoT solutions with the Dengue Patient Monitoring System (Team NEXORA).
- **Finalist** – Sri Lanka IoT Challenge (SLIoT) 2026: Developed the Dengue Patient Monitoring System (Team NEXORA).
- **Finalist** – Sri Lanka Robotics Challenge (SLRC) 2026.

## PROJECTS

**Dengue Patient Monitoring System (Team NEXORA) | *SLIoT & SPARK*** November 2025 - Present

- Engineered a wrist-worn ESP32 telemetry node to continuously measure and transmit three critical patient vitals (SpO2, blood pressure, and pulse pressure).
- Programmed I2C sensor protocols and established a robust 2-way serial communication link with a Raspberry Pi for centralized data aggregation.
- Deployed a lightweight machine learning model directly onto the hardware for localized, sub-second latency analysis of patient vitals.

### **Smart Water Management System | *Academic Group Project***

- Designed a custom PCB buck-boost converter and LDO circuit using Altium Designer to ensure stable 5V/3.3V logic power distribution.
- Developed a digital water flow tracking device integrating a pulse meter with Hall effect sensors and magnets to accurately monitor live water consumption.
- Interfaced a GSM module to wirelessly transmit 24/7 hardware telemetry to a cloud server, and produced comprehensive technical specifications and animated explainer guidelines.

### **Autonomous Robot Design (SLRC 2026) | *Competition Project*** July 2025 - April 2026

- Architected a computer vision pipeline capable of decoding visual payloads into precise grid coordinates.
- Compiled a custom C++ library to process a massive 56,000+ entry mapping table, successfully preventing 4GB RAM memory overflow crashes during live runs.
- Engineered a REST API bridge to sync real-time hardware telemetry and LED state indicators, achieving 100% system uptime during the competition.

### **Hardware Robot Design (Line & Wall Following) | *Academic Project***

Finished December 2025

- Constructed the physical robot chassis utilizing SolidWorks for precise mechanical part design and efficient hardware routing.
- Executed the complete physical assembly and strategic hardware integration, including the precise mounting of two sensors on the right side of the machine to facilitate the team's wall-following capabilities.

### **FPGA Co-Processor Accelerator | *Ongoing Academic Group Project***

Present

- Researching hardware architectures to design and implement a custom FPGA-based co-processor for computational acceleration utilizing a DE2-115 development board.
- Evaluating existing intellectual property (IP) cores and libraries for optimized computer arithmetic operations to integrate into the final accelerator design.

### **Digital Hardware Product Development | *Academic Project***

February 2025 - June 2025

- Executed the full hardware development lifecycle for a digital audio prototype, successfully progressing from custom PCB layout to the final physical enclosure design.

### **FPGA UART Communication Protocol | *Academic Project***

- Implemented a full-duplex UART serial communication protocol operating at a 115200 baud rate utilizing a DE0-Nano FPGA board.

### **Laboratory Database Management System | *Academic Group Project***

- Developed an administrative frontend dashboard interfacing seamlessly with a backend REST API to track and manage over 100 high-value laboratory assets.

## **POSITIONS OF RESPONSIBILITY & EXTRACURRICULAR**

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- **Committee Member, IEEE Robotics and Automation Society (RAS)** November 2025 - Present  
Co-organized the MAZEX 1.0 Micromouse robotics competition; drafted outreach communications and contacted professional mentors and industry resource persons for an associated online technical workshop series.
- **Organizing Committee Member:** ENTC Career Fair '26 (Jan 2026) & SLRC (Apr 2025)
- **Mentor, Soyuru Sathkara Seminar Series** January 2025  
Conducted targeted mathematics seminars for G.C.E. Ordinary Level (O/L) students to support their national examination preparation.

## **REFERENCES**

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### **Dr. Ranga Rodrigo**

Senior Lecturer, Dept. of Electronic & Telecommunication Engineering, University of Moratuwa

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